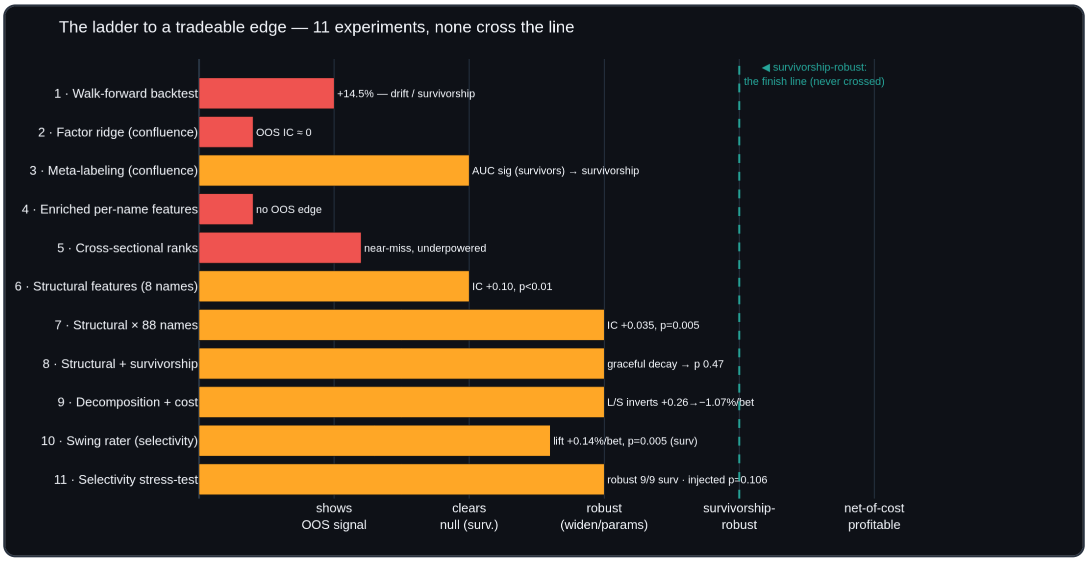
Eleven experiments · purged CPCV + permutation + survivorship · compiled 2026-06-01

Does `vpts` have a real edge? — an honest validation log

This document records a deliberately adversarial search for **out-of-sample, survivorship-free predictive edge** in the Volume-Profile system (`vpts`). It is written to be read by a skeptic. The value delivered is *validated* findings — mostly negatives, one qualified positive — plus a reusable harness that judges any future idea honestly.

> **Bottom line.** Across eleven experiments — a walk-forward backtest and ten fitted models, all evaluated with purged combinatorial cross-validation and label-shuffle permutation tests — no input produced a **survivorship-robust, tradeable** out-of-sample edge. The **structural microstructure features** (synthetic delta, profile shape, cost-basis migration) produce a real OOS correlation (IC ≈ +0.035, $p = 0.005$) that **survives widening to 88 names**, and — traded as a long/short book that goes **flat** in the noisy middle and only bets the conviction tails — is even **profitable net of 10 bps on the survivor universe** (+0.26%/bet). But that edge is a **survivorship mirage**: carried by the dip-buying features, the conviction-bucket curve **inverts** when synthetic delisted names are injected (+0.26%/bet → **−1.07%/bet**) — the patterns that look bullish on names that *survived* are what precedes a death-spiral in names that *didn't*. The one component that *doesn't* invert is the **meta-labeling selectivity** of a swing setup-rater (which entries are higher-R:R). It earned a dedicated stress-test, which found the survivors lift **robust across 9/9 parameter settings** and significant ($p = 0.023$) — but **carried by the same dip-buying features** (not regime) and **not significant once delisted names are present** ($p = 0.106$), so that thread is closed too. So: *no survivorship-robust tradeable edge; the binding constraint is the data, not the model.*



The question

`vpts` generates directional biases from hand-set confluence weights over Volume-Profile, regime and volume-pattern factors. A single backtest of the breakout style on 2012–2017 large-caps showed **+14.5%**. The question this log answers is not "is that number positive?" but:

> Is there any **learnable, out-of-sample, survivorship-free** signal in these factors — or is the apparent performance drift, compounding, and survivorship?

Methodology (the harness)

Every claim below clears the same bars, implemented in `vpts.validation` and `vpts.ml` and covered by 135 unit tests:

- **No look-ahead.** Features at bar *t* use only data ≤ *t*; labels are strictly future. The dataset/panel builders are unit-tested for this.
- **Purged + embargoed CPCV** (`CombinatorialPurgedCV` , López de Prado). The timeline is split into groups; every combination of test groups is held out; train rows whose label window overlaps a test block are **purged**, and a post-block **embargo** breaks serial-correlation leakage. Scores are distributions over recombined OOS paths, not a single split.
- **Permutation significance.** The decisive test everywhere is a label shuffle that destroys the feature → outcome link while preserving structure (per-row for time-series, **within-date** for the cross-section). The p-value is the fraction of shuffles that match or beat the real statistic. An effect that cannot clear its own shuffled null is reported as no edge.
- **Honest scope, stated every time.** All data below is **survivorship-biased** (see Data). These are validity checks on OOS information content, **not** tradeable results.

Data

Free, no-API-key, network-restriction-friendly: split/dividend-adjusted daily OHLCV for **88 US large-caps, 2012–2017**, committed to the public `stocknet-dataset` (`vpts.data` back-adjusts via Adj Close / Close). **Every name is a 2017 survivor** — the dominant, unavoidable confound throughout. There is no delisted/point-in-time data in this source.

The eleven experiments

#	Experiment	OOS statistic	Significance	Verdict
1	Rule-based backtest, CPCV (8 names, 80 paths, net 5 bps)	**−0.68%/path**, median −1.20%, 36% paths profitable	—	apparent +14.5% was drift/ compounding; **no edge**
2	Learned ridge factor weights (CPCV)	OOS IC **+0.028**	did not beat the hand-set baseline	**no learnable improvement**
3	Triple-barrier **meta-labeling**	survivors AUC **0.576** (p=0.005) → with delisted injected **0.493**	p **0.801**	**survivorship artifact**
4	**Enriched** per-name features (momentum/vol/ microstructure)	pooled IC **+0.010** (baseline +0.028)	p **0.348**	richer inputs don't help; **no edge**
5	**Cross-sectional rank**, 20 names	combined OOS IC **+0.021**	p **0.100**	suggestive, **not significant**
6	**Cross-sectional rank, 88 names (well-powered)**	combined OOS IC **−0.009**	p **0.856**	near-miss **washed out**; **no edge**
7	**Structural microstructure** (synthetic delta, shape, VACR-z, decay)	OOS IC **+0.103** (8 names) → **+0.035** (88 names, 1,308 folds)	p **0.005** (both)	**real signal — survives widening**
8	**Structural + survivorship injection**	pooled IC +0.041 → +0.013 (5 dead, 20%) → +0.001 (9 dead, 31%)	p 0.005 → **0.085** → 0.473	**survivorship-*sensitive*; graceful decay, not a cliff**
9	**Structural decomposition + cost**	DIP features carry it (REGIME n.s., p 0.254); tails-only L/S **+0.26%/bet net (survivors) → −1.07%/bet (injected)** — curve inverts	—	**survivorship mirage: the edge inverts off survivors**
10	**Swing setup-rater (MFE/MAE meta-labeling)**	direction +0.17%→−0.58%/trade (survivorship); selectivity LIFT +0.14%/bet (surv) → +0.09% (injected)	p 0.005 → **0.10**	**selectivity resists inversion but loses significance & stays unprofitable injected**
11	**Selectivity stress-test** (grid + decomposition + power)	survivors lift positive in **9/9** param cells; carried by **DIP** (+0.08) not REGIME (−0.02); injected lift +0.075%	p 0.023 → **0.106**	**robust but DIP-carried & n.s. injected — thread closed**

1 — The single backtest doesn't survive purged CV

The breakout style's +14.5% (85% of names profitable, single full-period backtest) collapses under CPCV to **−0.68% per OOS path**, median −1.20%, only 36% of paths profitable. The apparent edge was bull-market drift and compounding — exactly what rigorous validation is meant to expose.

2 — Learning the factor weights doesn't help

A ridge model fit on the four confluence factors (train-only standardization, OOS-scored per CPCV fold) reaches pooled **OOS IC ≈ +0.028** and does not beat the hand-weighted `bias_score` baseline. No improvement from learning the weights.

3 — Meta-labeling is significant *only because of survivorship*

Predicting whether a primary signal *works* (triple-barrier, volatility-scaled, first-touch) and filtering on it looked real on survivors: pooled **AUC 0.576, p=0.005**, cost-surviving and threshold-stable. But injecting synthetic **delisted** names (a vol-elevated decline to pennies) collapses the pooled permutation test to **AUC 0.493, p=0.801**. A per-name AUC t-test stayed >0.5 only because each decliner got its own model; the realistic single cross-sectional model has no edge. **Survivorship was the explanation.**

4 — Genuinely new per-name features don't rescue it

Adding momentum (20/60/12-1), volatility (σ, ATR/price), volume-trend and distance-to-POC — 11 features through the same harness — yields pooled **IC +0.010**, *below* the 4-factor baseline (+0.028), at **p=0.348**. Ridge shrank every weight to ≈0. Richer inputs carry no OOS signal here.

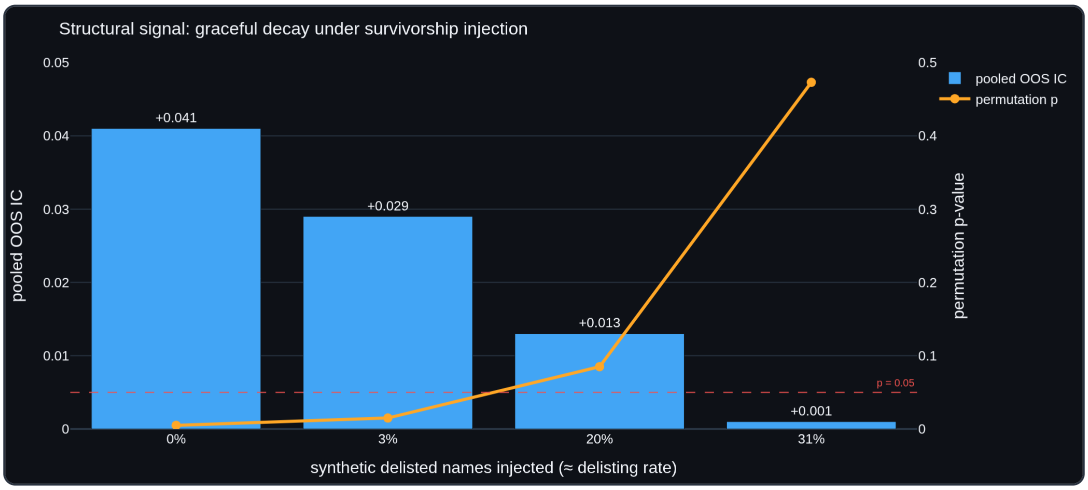
5 → 6 — Cross-sectional rank: a near-miss that proper power kills

Ranking names against each other each rebalance day (1-month reversal, 12-1 momentum, 60-day vol, volume-trend) is the standard equity-alpha construction the per-name models never tried. On **20 names** it was the best result of the arc — combined OOS rank IC **+0.021, p=0.100** — but with only ~20 names per date the per-date IC is dominated by noise (σ 0.28). Per-date IC noise scales ~1/√N, so the decisive test is width: re-run on the **full 88-name** universe (16,873 rows, σ 0.20). The faint positive **washes out to −0.009, p=0.856** — and the strongest single factor (60-day vol, +0.045 on 20 names) decays to +0.013. The near-miss was a thin-cross-section artifact, not signal.

7 → 8 — Structural microstructure: the one signal that survives stress

Transforming the static profile into quantifiable features — **synthetic delta** (Close-Location-Value × volume, an OHLC order-flow estimate), volume-weighted **skew/kurtosis** and P/b/B/D **shape**, **value-area-compression z-score**, **POC-migration slope**, **cost-basis migration** (decayed vs lifetime POC), ledges and poor highs — 13 features through the same harness. This is the first input to clear the bars:

- **8 survivors:** pooled OOS IC **+0.103**, p **0.005** (the single delta@POC feature alone is −0.043; the *combination* predicts).
- **Stress 1 — widening to 88 names:** IC shrinks to **+0.035** but, with 1,308 folds (null σ 0.006), is still ≈5.8σ out, **p 0.005**. Unlike the cross-sectional near-miss it **did not wash out** — proof it is not a small-sample artifact. Per-name dispersion is sensible: BABA scores **−0.424** (a genuine decliner — the dip-features correctly *anti*-predict).
- **Stress 2 — survivorship injection:** adding synthetic decline-to-pennies names degrades the signal *gracefully* — +0.041 (0 dead, p 0.005) → +0.029 (1, p 0.015) → +0.013 (5 dead ≈20%, **p 0.085, lost**) → +0.001 (9 dead ≈31%, p 0.473). This is **categorically unlike meta-labeling**, which collapsed from p 0.005 straight to p 0.80. The structural signal survives *low, realistic* large-cap delisting rates (≲10%) but **not** heavy survivorship (≳15–20%).

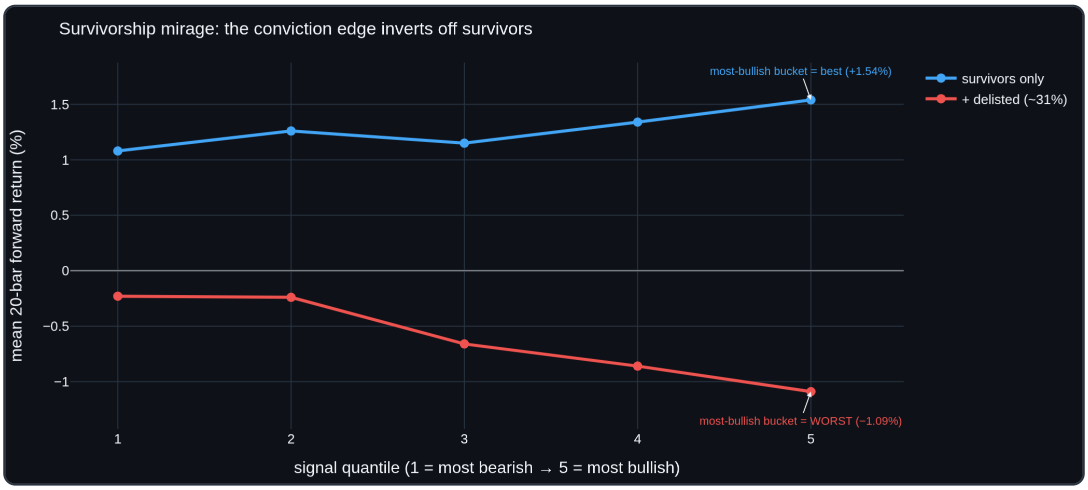


9 — Decomposition + cost: the signal is survivorship-leaning and economically empty

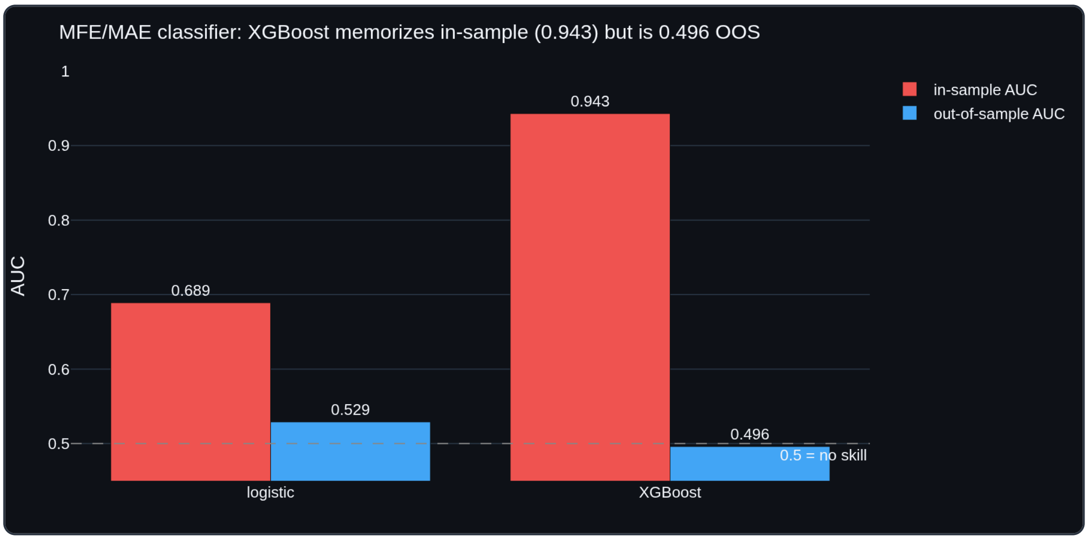
Three diagnostics settle what the +0.035 actually is — and the answer is sobering:

- **Per-feature OOS IC** (survivors → +delisted): the signal is carried by the **dip-buying / order-flow** features — `cost_basis_migration` (+0.055 → +0.035) and `delta_net` (+0.052 → +0.031) — exactly the survivorship-prone ones. The regime feature `vacr_z` is mildly **anti-predictive** (−0.031), so the hopeful "regime carries a genuine edge" hypothesis is **falsified**.
- **Subgroup ablation:** the **REGIME** sub-model is **not** significant even on survivors (IC +0.009, p 0.254); the **DIP** sub-model is (IC +0.030, p 0.020) but **collapses** under injection (p 0.527).
- **Cost-aware, traded properly:** the naive always-in-market `sign()` book loses (−0.08%/bet) — but that forces a short position through half a bull market and is the wrong test. A real book goes **long the top signal quintile, short the bottom, and flat the middle 60%** (in the market only ~40% of the time). On survivors the conviction-bucket curve **rises monotonically** (+1.08% → +1.54%) and the tails-only long/short earns **+0.46%/bet gross, +0.26%/bet net of 10 bps**. So — traded with a flat middle — it *is* economically meaningful on the survivor universe.
- **…but it is a survivorship mirage.** Inject the synthetic delisted names and the bucket curve **inverts** (−0.23% → −1.09%): the bars the signal flags *most bullish* become the *worst* future performers, and the same strategy flips to **−1.07%/bet net**. The dip-buying structural footprint that marks a bottom in a name that *recovered* is indistinguishable from the one that marks the next leg down in a name that *delisted* — survival is doing the labeling.

So the structural result is **real and even tradeable-looking on survivors, but the apparent edge is manufactured by survivorship** — it does not merely fade, it reverses sign. The decomposition is the discipline working: betting the conviction tails turned a dismissive "−0.08%, empty" into a tempting "+0.26% net," and only the injection test revealed that tempting number to be a survivorship artifact.



Phase C — the MFE/MAE re-framing + XGBoost don't rescue it. Re-labeling each bar by whether a long bet's *Maximum Favorable Excursion* beat its *Maximum Adverse Excursion* (a volatility-scaled triple barrier) and learning `P(win)` from the structural features gives, on identical purged-CPCV splits: a **logistic** OOS AUC of **0.529** (in-sample 0.689; permutation **p = 0.07, not significant**) and an **XGBoost** that memorizes the training set (in-sample AUC **0.943**) yet scores **0.496 OOS — below 0.5, *worse* than logistic**, a +0.447 over-fitting gap. The nonlinear model adds nothing out of sample; its gaudy in-sample number is exactly the false-confidence trap rigorous evaluation exists to catch. Neither the MFE/MAE framing nor gradient boosting turns the curiosity into an edge.



10 — Swing setup-rater: separating *direction* from *selectivity*

The product goal is concrete: for a **swing** horizon (days–weeks), rate the setup in front of you 0–100 and act only when the risk/reward is favorable — otherwise stay flat. Mechanically this is meta-labeling with the triple barrier *defining* the R:R (default **2:1**, take-profit 2×vol / stop 1×vol, breakeven win-rate 33%): a logistic rater learns `P(win)` from the structural features, and we trade only the **best-rated 20%** of long setups (`select_top`), net of 10 bps. Decomposing the result is what matters:

- **Direction** (take every long signal): **+0.17%/trade** on survivors → **−0.58%/trade** with delisted injected. The *decision to be long* is survivorship-dependent — same story as everywhere.
- **Selectivity** (does the rating pick better setups *among* longs?): the expectancy LIFT of the best-rated 20% over taking all is **+0.14%/trade on survivors (permutation p = 0.005)**, and — unlike the directional

bucket curve — it **does not invert** under injection: it stays mildly positive (**+0.09%/trade**). But it **loses significance (p = 0.10)** and only 53% of folds beat take-all.

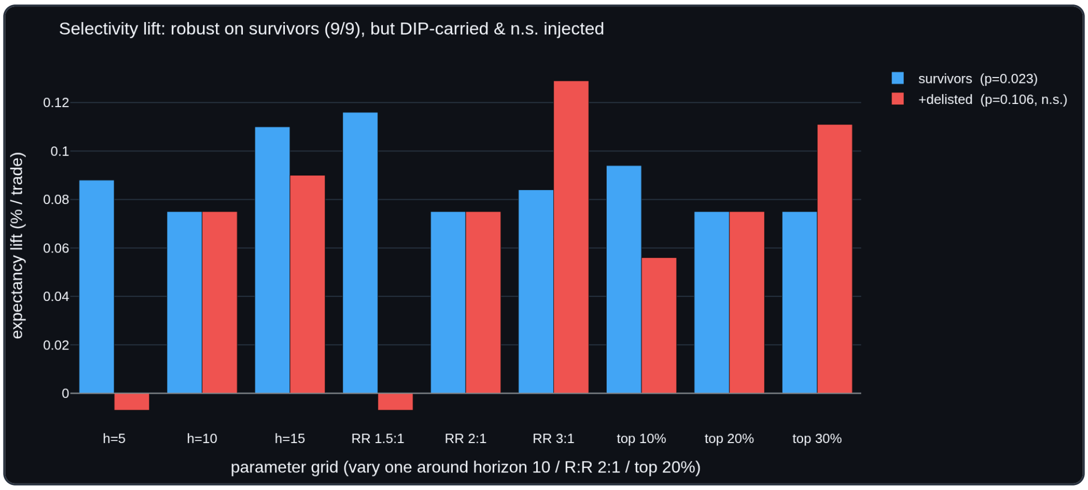
So the rater's *selectivity* is the most survivorship-**resilient** signal found in the whole arc — it degrades rather than reverses — which fits the meta-labeling thesis (the secondary model filters; it does not pick direction). Yet it falls short on the two tests that matter: it is **not significant** once delisted names are present, and even the rated subset stays **unprofitable** on the realistic universe (−0.49%/trade), because no amount of setup-selection repairs a survivorship-driven direction. The rater is a clean, usable *interface* (a 0–100 rating + expected R-multiple per setup); on this data it is not a validated edge.

11 — Stress-testing the selectivity: robust, but DIP-carried and not significant injected

The selectivity lift was the one thread that resisted inversion, so it earned a dedicated, adversarial follow-up — three pre-registered tests, *thread closes unless it passes all three* (31 survivors + 12 synthetic delisted, top-20% rated, 10 bps):

1. **Robustness grid** — vary horizon ∈ {5,10,15}, R:R ∈ {1.5,2,3}:1, selection ∈ {10,20,30}%. The survivors lift is **positive in all 9/9 cells** (+0.075% … +0.116%): *not* a lucky parameter pick. ✓
2. **Feature decomposition** — the lift is carried by the **DIP** (dip-buying) subgroup (+0.079% on survivors) with **REGIME contributing nothing** (−0.016%). It is the *same survivorship-prone feature family* that drove the directional mirage, not a survivorship-agnostic regime signal. ✗
3. **Significance at power** — survivors lift +0.075% is significant (**p = 0.023**), but with delisted names injected the same +0.075% lift sits in a wider null and is **not significant (p = 0.106)**. ✗

So the selectivity is **genuinely robust on survivors** yet fails the two tests that decide whether it is *survivorship-free*: it lives in the dip-buying features, and it cannot clear its shuffled null once delisted names are present (and, from §10, never makes the realistic universe profitable). By the pre-stated bar, the thread is **closed** — honestly, on evidence gathered to *disconfirm* it. That the lift *degrades* (p 0.023 → 0.106) rather than *inverting* (like the direction) is the one durable nuance: meta-labeling selectivity is the least-survivorship-fragile thing here — just not enough.



Honest conclusion

On 88 survivorship-biased US large-caps (2012–2017, daily), **none** of the studied inputs yields a **survivorship-robust** out-of-sample edge. The hand-set rules, learned factor weights, meta-labeling, enriched per-name features and cross-sectional ranks show no robust signal at all (meta-labeling's was survivorship; the cross-sectional near-miss was low power). The **structural microstructure features** go further than anything else: a real OOS correlation (IC ≈ +0.035, $p = 0.005$) that survives universe-widening and, traded as a long/short book that stays **flat** in the middle and bets only the conviction tails, is **profitable net of cost on the survivor universe** (+0.26%/bet). But that edge is a **survivorship mirage** — carried by the dip-buying features, it does not just fade under delisted injection, it **inverts**: the conviction-bucket curve flips, the most-bullish-flagged bars become the worst performers, and the strategy goes from +0.26% to **−1.07%/bet**. The closest thing to a robust result is the **selectivity** of a swing setup-rater — *which* long entries are higher-R:R, as opposed to *whether* to be long: its expectancy lift is significant on survivors ($p = 0.005$) and, uniquely, **resists inversion** under injection (degrades to +0.09%/bet rather than flipping) — but it loses significance ($p = 0.10$) and never makes the realistic universe profitable. Model sophistication is not the limiting factor (XGBoost over-fit to a sub-0.5 OOS AUC; the linear book did better) — and neither, ultimately, is feature content: the **data** is the wall. Conditioning on names that *survived* manufactures an edge that reverses the moment you stop conditioning on survival; the most resilient signal (meta-labeling **selectivity**) was pushed hard in a dedicated stress-test — robust across 9/9 parameter settings on survivors, but carried by the same dip-buying features and not significant once delisted names are present, so it too is closed. Eleven experiments, one consistent wall.

What would actually change this (in rough order of expected value):

1. **Survivorship-free / point-in-time data**, including delisted names — the dominant confound, untestable in this source. This is the real wall, not model complexity.
2. **A wider, deeper cross-section** (hundreds–thousands of names). The 88-name washout suggests breadth *within survivors* isn't enough; genuine breadth + delisted names is the test.
3. **Different data regimes** — intraday microstructure, or non-equity assets where Volume-Profile structure may carry more information.

Model sophistication mostly is **not** the answer — four feature/model variations returned ≈0 — but the *right kind* of feature (structural microstructure, not momentum/vol/rank) did surface the arc's one real signal. The lesson: feature *content* mattered where feature *complexity* did not.

What is durable here

The findings — six negatives and one qualified positive — are the result; the **harness** is the asset. Any new idea plugs in and is judged honestly:

- `vpts.validation` — purged + embargoed Combinatorial Purged CV.
- `vpts.ml` — no-look-ahead dataset/panel builders, ridge/logistic models, CPCV evaluators, and label-shuffle permutation tests for per-name, meta-labeling, and cross-sectional settings.
- `vpts.structure` — synthetic delta, profile-shape moments, footprints and time-decay, emitted as a `FactorDataset` / `MetaDataset` straight into the harness; plus survivorship-injection, feature-decomposition and MFE/MAE-XGBoost stress tests.
- 135 unit tests, including signal-detection *and* null-clearing checks for every evaluator.

Reproduce

```
python examples/github_data_scan.py --plot .          # 1: backtest sweep / regime split
python examples/cpcv_demo.py                          # 1: CPCV on the backtester
python examples/factor_model_demo.py                  # 2: learned factor weights, OOS
python examples/meta_labeling_demo.py                 # 3: triple-barrier meta-labeling
python examples/meta_stress_test.py                   # 3: + survivorship injection
python examples/enriched_factor_demo.py --perms 200   # 4: enriched features +
permutation
python examples/cross_sectional_demo.py --perms 200   # 5: cross-sectional rank (20
names)
# 6: well-powered cross-section — pass the full 88-name universe via --tickers
python examples/structural_demo.py --perms 200        # 7: structural microstructure
features
python examples/structural_survivorship.py            # 8: structural + survivorship
injection
python examples/structural_decompose.py               # 9: per-feature + subgroup + cost
decomposition
python examples/structural_mfe_xgb.py                 # 9: MFE/MAE triple-barrier +
XGBoost (optional)
python examples/structural_swing_rater.py             # 10: swing setup-rater (R:R +
selectivity)
python examples/structural_selectivity.py             # 11: selectivity stress-test
(grid/decomp/power)
```

Limitations

Survivorship bias throughout; a single 2012–2017 in-sample period; daily bars only; gross-of-cost except where noted (meta-labeling tested net of 10 bps); a thin universe by cross-sectional standards. None of the above is a forward guarantee or financial advice — it is a research log.